

SI 500: MILESTONE 5 - Final Report & Presentation

Team 2:

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SI 500

Section 011

Partner: Instructure

Partner Summary

Instructure's mission is to help learners thrive in tomorrow's landscape by delivering a future-ready ecosystem. The company provides leading EdTech products, including Canvas, a learning management system; Mastery, a K-12 assessment platform; and Parchment, a credential and record management system.

Problem statement and HMW question

How might Instructure refine its current products or develop a new product to include individuals on alternative pathways into its educational ecosystem, empowering them by strengthening lifelong learning and employability, therefore ensuring Instructure's long-term competitiveness in the future EdTech Industry.

Word Count:

Wentao He

Table of Contents:

- **Table of Contents:** Section titles and starting page numbers. Wentao He Ping Yin
Not Started ▾

Executive Summary:

- **Executive Summary:** One-page overview of key points, findings, and recommendations
Wentao He Ping Yin Not Started ▾

Introduction: Completed ▾

The landscape of higher education in the United States is undergoing a profound transformation. Traditional four-year college pathways are increasingly viewed as expensive, time-consuming, and misaligned with labor-market demands, leading 79% of Americans to opt out of attending traditional universities (Nietzel, 2025; Emmons et al., 2019). This shift has accelerated the rise of alternative education pathways, bootcamps, and online learning platforms, which now serve 61% of learners (National Student Clearinghouse, 2025). These alternatives are further propelled by technological advancements, particularly AI, which has shortened the gap between learning and doing, enabling more flexible, personalized, and skill-based learning experiences (Kestin et al., 2025).

Despite their growing popularity, alternative education models face significant structural challenges. Many lack standardized learning roadmaps, reliable assessment tools, and recognized certifications, which hinders learners' ability to demonstrate their skills to employers. As a result, 29% of individuals report being negatively impacted by the absence of a dependable learning platform (Nietzel, 2025). While employers increasingly embrace skill-based hiring, degree requirements still dominate recruitment processes, creating barriers for those without formal credentials (Carnevale et al., 2024). These issues are compounded by deep information gaps between learners, employers, and educational institutions, which prevent effective alignment between learning outcomes and workforce needs.

This context presents both a challenge and an opportunity for EdTech companies like Instructure. Currently, Instructure's platforms, such as Canvas and Mastery, primarily serve traditional institutions, limiting their reach among independent and marginalized learners. Competitors like D2L Brightspace have already begun expanding into non-traditional learning communities, positioning themselves to capture a broader share of the evolving education market (D2L, 2025). To remain competitive and socially impactful, Instructure must rethink its approach and expand its ecosystem to include learners on alternative pathways.

Our team's project addresses this critical gap by exploring how Instructure can refine its existing products or develop new solutions that validate skills, provide structured learning pathways, and enhance employability for non-traditional learners. This includes integrating modular learning roadmaps, competency-based assessments, and portable digital credentials that are recognized by employers. By doing so, Instructure can empower individuals to pursue lifelong learning and career mobility, regardless of their educational background.

The relevance of this challenge extends beyond product innovation it touches on equity, access, and the future of work. As the demand for flexible, skills-based education grows, EdTech platforms must evolve to support diverse learners and bridge the gap between education and employment. Instructure is well-positioned to lead this transformation by leveraging its technological capabilities and institutional partnerships to create inclusive, scalable, and impactful learning solutions. Addressing this problem not only ensures Instructure's long-term competitiveness but also contributes to a more equitable and responsive educational ecosystem.

Problem Statement Completed ▾

Our project focuses on bridging the gap between traditional education, which is often costly, time-intensive, and produces uneven outcomes across different backgrounds, and current alternative pathways that, while more flexible, frequently lack credibility, structured learning roadmaps, and tools. The scope includes exploring how Instructure can refine its existing platforms, such as Canvas and Mastery, or develop new solutions that support learners pursuing non-traditional routes, including bootcamps and self-paced online learning. Based on our research, we recommend integrating structured pathways and robust AI learning mechanisms. This approach ensures that Instructure remains competitive in the evolving EdTech landscape by addressing the shortcomings of both traditional and alternative education. The guiding question is: *How might Instructure design or adapt products to validate skills, provide structured pathways, and strengthen employability for learners outside traditional education, thereby securing its long-term role in the future of EdTech?*

Process/Methods Completed ▾ Wentao He

To ensure that our recommendations are based on real-world issues and reality, we followed these rigorous stages of research.

First, the background research report. After Instructure's presentation, our team decided to focus on investigating how Instructure should respond to the impact of the current higher education system and the growing alternative education pathway. Based on this target, our research covered different aspects of the current situation, and we can analyze how the development of society, technology, and economy can have a huge impact on people's way of learning, leading to huge societal shifts in learning pathways. Besides, we benchmarked Instructure's current products against its competitors, including other LMS platforms such as D2L Brightspace, and other alternative education platforms, such as Coursera. Finally, we identified the gap in Instructure's Ecosystem, including features offered by competitors that Instructure lacks, market needs that Instructure has failed to meet, and critical missing capabilities hindering Instructure's achievement of its core goals.

Based on background research, we created our Live Project Plan. In the live project plan, we summarized the background context of our project, in which we identified the difficulties facing traditional higher education, the significant market gaps and structural barriers in alternative education pathways, and Instructure's challenges (competitors like D2L Brightspace have already gained an early foothold in the alternative education pathway market) and opportunities (the market needs reform and Instructure has technological foundation). We then identified the stakeholders and projected the expected Project Outcomes, an actionable recommendation, and the project's expected deliverables, including prototypes and reports. Finally, we divided the projected tasks to ensure effective collaboration within the team.

Next, we proceeded to the Data Collection phase. Our data source includes interviews, institutional research, and business journalism. For the interview, initially, our participants included three former alternative education pathway students currently in the workforce, two HR

professionals, and one Instructure employee. After designing our protocols, conducting the interviews, and submitting our memos, we received feedback indicating that because the two former students we interviewed were both Chinese, their experiences reflected a social environment distinct from that of the United States. Therefore, we adjusted our plan and recruited an American alternative education pathway student. The suggestion proved right; while there are parallels between the experiences of Chinese and American students, significant differences remain. By interviewing this new participant, we successfully bridged the data gap regarding personal experiences within the American alternative education context.

It is worth noting that a major change in our understanding of the problem during the interviews concerned the collective defined by 'alternative education pathway students.' In our earliest design, we focused solely on the 'next steps' for high school graduates who chose not to attend a traditional four-year university. However, based on the interviews, we discovered that learning is often a lifelong endeavor. What's important is not just the 'first job,' but continuous learning. It is common for alternative education pathway students to discover skill gaps that need addressing while working or to desire a career change. Therefore, our design must account for the much larger group of 'in-career alternative education pathway students,' rather than assuming they have all just left high school.

After data collection, we used affinity diagrams as our primary analysis tool. This was an incredibly difficult process; sorting hundreds of notes was a huge engineering effort. We struggled to balance the size of the smallest clusters while synthesizing them into high-level themes, often amidst chaotic discussions. Yet, this became the phase where we derived our most significant findings and insights. In the process of designing the Affinity Wall, team members must focus on the natural relationships and similarities between the notes, using the raw findings from the interviews rather than their own ideas for clustering and refinement. To allow the resulting clusters to be grouped into higher-level clusters, we reorganized the initial clusters and repeated the cycle several times. This iterative process allowed us to gain a deeper understanding of the collected data.

Findings/Insights Completed ▾ Wentao He

Note: We categorize our Interview Notes by Stakeholder groups. When listing the original interview notes below, 'A' stands for Alternative Education Pathway Student, 'I' stands for Instructure Employee, and 'H' stands for HR.

Structural Incompatibility: For many Alternative Education Pathway Students, a significant difference from Traditional College Students is that they cannot study full-time. This group primarily consists of adults who have already entered the workforce. After securing their first job, they realize they need more knowledge but cannot study full-time like university students. Their time is fragmented—it might be 20 minutes on the subway commute, or half an hour at night after the children are asleep. If course videos strictly follow the university's open lecture format of lasting forty-five minutes, it is difficult to get started. Therefore, our solution must be fragmented, capable of linking knowledge into units lasting a few minutes to adapt to this pattern.

Data:

- I: “Non-traditional students may have other things going on in their lives.”
- A: “(While I was learning), I worked half-time during the day, so it was Monday, Wednesday, and every other Friday, and then I worked 3 nights a week.”
- A: “I just need a list of instructions that I can follow.”
- A: “People who do videos, uh... usually I have to speed them up.”

Need-Driven Engagement: For adult learners who have forgone the traditional four-year college in favor of an Alternative Education Pathway, their learning is often utilitarian, pragmatic, and driven by a sense of urgency. Traditional college education often builds a solid foundation and accumulates knowledge to construct a complete intellectual framework, or even serves to cultivate character. However, while much of this knowledge is necessary, it is not urgent; alternative education pathway students generally tend to learn urgent knowledge first. Therefore, the learning material we provide must be reconstructed with the specific goal of “application” in mind.

- A: “I hate the format. It's like... these modules of click-throughs to different systems that don't talk to each other. And it doesn't make any sense, like... It's not streamlined or logical to me.” (Complaining that the way Canvas LMS's module provides the knowledge does not meet her needs)
- A: “I knew if I was going to start a business, I would need to figure out accounting, so ... I went and took an accounting class.”
- A: “I don't need a bunch of, like, stories about how to solve a problem. ... I just need the problem. And information about what makes sense, and then how to do it.”

Mid-Journey Disorientation: Mid-Journey Disorientation is a bottleneck very easily encountered by self-learners. In traditional education, people at least know what grade they are in, which courses they have taken, and how far they are from graduation requirements; however, self-learners often find themselves in an endless intermediate state. In school, one can judge whether they are ahead or behind based on the progress of classmates, and the degree itself is a target, but in self-learning, it is easy to feel lost. Besides, alternative education pathway students often move between various bootcamps or online course platforms to learn different techniques. The resulting consequence is highly fragmented knowledge. Such fragmentation not only hinders the construction of a systematic knowledge graph but also makes students feel insecure, leaving them unconfident about the use of what they have learned and how to compete with those holding university degrees.

- A: “The knowledge gained through self-learning was highly fragmented and lacked a systematic structure.”
- A: “(When I need the knowledge), previously acquired knowledge has been forgotten, rendering the process of relearning more challenging.”
- I: “Non-traditional students may not coming for a degree”

The Employability Gap: For those looking for their first job or wanting to switch careers, the most practical result an alternative education pathway can bring is a job offer. The labor market has already established a sense of trust in traditional university degrees, whereas providers of alternative education pathways have not been able to offer an equivalent credential. For many people currently on alternative education pathways, learning is pursued for its own sake, and certificates are often viewed as mere keepsakes. However, if these credentials carried real market

value, they would certainly boost their job prospects. Beyond valuable certifications, if alternative education platforms could connect directly with employers to provide referrals for relevant internships, that would offer an even more tangible benefit.

- A: “If I can tell people that I’m, for example, ICPI certified... They have more respect for me.”
- A: “My second job was secured through a referral from a secondary school classmate.”
- A: “The certificate of completion itself was not helpful, since it did not carry much weight or credibility for employers.”
- A: “Culinary schools like New East Cuisine Education offer targeted internships at restaurants. Outstanding students who become chefs can also promote New Oriental itself.”
- I: “Credentialing is only good if it has currency in the job market. Traditional degree itself has currency, and people know how to value it.”
- I: “Industry hasn’t figured out the value behind the certificates.”

Soft Skill Barrier: Traditional Education places great emphasis on the cultivation of Soft Skills. Undergraduates are required to join group projects to learn how to collaborate with colleagues; there are also very practical soft skill training exercises, like simulation interviews and presentations. The cultivation of soft skills is dispersed throughout the four-year curriculum. However, existing Alternative Education Pathways often pursue efficiency and quantifiable outputs, tilting the teaching focus toward hard skills that can help build a portfolio or directly meet recruitment requirements. This leads to learners being in a closed state of "working in isolation" for a long time, severely lacking training in teamwork, non-violent communication, and dealing with ambiguous requirements—all of which are essential in the real workplace. To avoid producing students who can solve interview questions but appear helpless when facing complex interpersonal feedback or cross-departmental collaboration, the alternative education pathway must also provide a method to cultivate people's soft skills.

- I: “The non-traditional education misses out on soft skills.”
- A: “It (the alternative pathway learning) can be very isolating.”
- A: “The social support probably would have made things much better.”
- A: “I got promoted not because of any special technique, but the boss saw me working hard.”

Recommendations & Next Steps

Primary target users

Our primary target users are individuals aged 18 to 35, who are currently pursuing alternative learning pathways. Our target users must work and study simultaneously. They are generally self-motivated and clear about what they want to learn. However, they do not have sufficient time to study. Furthermore, they may have different learning habits: learning is not their first priority, so they want a more straightforward way of learning. Another frustration they face is that, due to the usual degree-based screening in the hiring process, they often need to effectively communicate their experiences and soft skills to employers to demonstrate their competencies, whereas traditional learners are naturally assumed to be equipped with such soft skills.

To better understand these users, we created a persona that highlights their major pain points: limited study time, lack of guidance in learning, and difficulty translating their experience into employable competencies.

Primary Persona

"I'm done with scattered learning. I want a clear path that actually makes me employable without a formal degree"



Lauren Case

Age: 29

Pathway: Alternative, formal school early, pursued work + self-learning

Hometown: Southfield, MI

Education: Participates in short-term online bootcamps.

Occupation: Operations coordinator at a small company

Pain Points

- **Limited time in studying**
- **Lack guidance** in learning skills
- **Soft-skill gaps** aren't addressed by bootcamps.
- **Scattered learning**
- **Limited access** to knowledgeable mentors.
- Difficulty **translating hustle into employable competencies**.

Motivations

- Driven by embarrassment she once felt working with **more educated colleagues**.
- Wants **fair access** to jobs without needing a degree.
- Desires a sense of **competence and systematic thinking** in her work.
- Believes deeply in **constant learning** and wants **guidance**, not random self-learning.

Goals

- Build **structured knowledge** rather than fragmented learning.
- Gain **practical soft skills** (communication, teamwork, logic, problem-solving).
- Become more **"officially qualified"** through meaningful micro-credentials.
- Access a pathway that connects learning to **real employability**.

<Figure 1. Primary Persona>

Recommendation 1: Personalized Modular Learning Platform

• Connection with Research Findings

The first key pain points we would like to address for the users are the lack of sufficient time for a full learning session. Our interview participant A3 mentioned that "I find watching a video to be irritating, as when I'm learning how to do something...I have to speed them up by one and a half or two times the amount, because I can't stand how slow they are...Also, many instructions can be too long and too verbose...I have to work, and I don't have time for this". This reveals one of the most challenging aspects of their study process. Despite having the motivation for self-directed study, they allocate the majority of their time to other aspects of their life: work, family, or other responsibilities. In other words, studying may not be their top priority, and traditional lectures either drain their energy or are simply impossible to manage. Therefore, our recommendation aims to accommodate their busy schedules by providing a flexible, smaller chunk of learning materials.

• The Recommendation

Our first recommendation is a personalized, modular learning platform. It is a knowledge-focused learning platform that can accommodate different learning styles, offering modularized, flexible learning experiences. The main component of the platform is a feed similar to TikTok-style short content, but focused on educational micro-content, such as short videos explaining a concept, tiny knowledge cards, or quick tips related to the user's interests. The interface is light, browsable, and modularized, which ideally suits a 10- to 20-minute lunch break or waiting time.

The platform personalizes the feed based on the user's searches and recommendation algorithm, which reduces the user's cognitive load: "users do not need to search extensively with the

relevant micro-modules surfacing automatically. Additionally, this modular learning platform enhances flexibility: It supports unplanned or curiosity-based learning behaviors, and it gives users a low-pressure, flexible starting point to learn something new. This feature refers back to our interview participant A3's suggestions of “a lower-stress way to keep going and gaining the skills needed”. If users are on a quick 10-minute break and are curious about terminology in personal finance, they can type keywords such as 'personal finance' into the search bar and receive immediate suggestions with bite-sized knowledge, rather than navigating through verbose instructions.

<Visual Mockup 1.  Personalized Modular Learning Platform Demo.mov >

- **Why It Matters**

This recommendation is tailored to the primary needs of the target user; it also utilizes Instructure’s business competencies of having the largest LMS and related resources. It makes a high impact on learner experience and access for non-traditional students by offering flexible, modular, knowledge-oriented pathways instead of rigid coursework.

This recommendation is also scalable and has high potential to open new revenue streams (subscription access, micro-credentials, modular programs) and expand Instructure’s market reach, while also being innovative and socially responsible. It serves as a strategic move toward lifelong and non-traditional learners. Even though it may not be required for survival now, it is important for long-term competitiveness.

In all, it is highly aligned with learner needs and leverages Instructure’s existing ecosystem.**[Add citation from Jody]**

- **Implementations and Constraints**

The recommendations require content partnerships or a modular content taxonomy **[Cite Jody’s feedback]**. Many universities are providing their online learning resources through online learning platforms such as Coursera or Udemy, while also developing their own platform, such as Michigan Online. We suggest Instructure partner with these online learning platforms. The quality cannot only be maintained by the online platform partner and Instructure, but also can be assessed by the user using votes and reviews similar to streaming platforms. Time-wise, the recommendation may require a medium to long time to plan and implement, because it requires content modularization, recommendation logic, and UX changes. The initial cost could be high (content rework, algorithm, and engineering), but spread over a large user base and multiple programs, so the investment can be justified.

Recommendation 2: Skill Tree Roadmap for Learning Progress

- **Connections with Research Findings**

Another pain point of our primary target user is the lack of clear guidance on the whole learning process. Compared to traditional students taking full-time courses with clear syllabi and defined learning expectations, non-traditional students often face a dilemma. They know what they need to learn, but when they start exploring a concept, they realize they lack certain background knowledge or are unaware of related concepts. Our interview participants A1 and A2 both mentioned, “a challenge in learning stems from a lack of prior understanding.”

- **The Recommendation**

Our second recommendation is a skill tree roadmap that visualizes users' learning progress and provides clear guidance throughout the learning process. This skill tree gives users a simple overview of what they have already covered, what background knowledge is required, what related concepts exist, and what modules are recommended next. This recommendation helps learners who feel unsure about learning direction or scope stay on track with their learning progress, while also supporting the transition between curiosity-based learning to more intentional learning. It is designed to be lightweight and easily integrated into Canvas as a helpful, portable guide.

<Visual Mockup 2.  Skill Tree Learning Roadmap Demo.mov >

- **Why It Matters**

The recommendation addresses a clear learner need. By showing prerequisites and progress in a tree format, it improves clarity, motivation, and self-direction for learners, enabling them to navigate learning without feeling overwhelmed.

This is the most feasible of the three recommendations. It builds on existing tools such as outcomes, credentials, and pathways, and fits naturally inside Canvas and can be adopted incrementally. [Cite Jody] The feature itself is portable and value-adding which can support retention and satisfaction. It can also integrate with the modular learning platform and function as a coherent “learning navigation system.”

- **Implementation and Constraints**

The recommendation requires less time to implement than the other two, since it can be layered on top of existing course, module, and outcome data as a visualization feature. Additionally, implementation costs mainly depend on UX, data modeling, and integration work; it does not require reworking the entire platform. One constraint of this recommendation is that institutions vary widely in how structured their program outcomes are. Hence, a template that is adaptable and flexible across institutions and courses could perform better than a rigid “study outline”[Cite Jody]. Therefore, the recommendation should be designed to provide users with a high-level guideline of the learning process and support user navigation without overwhelming them with fixed requirements.

Recommendation 3 - Soft-skill Development Tool Ping Yin In Progress ▾

- **Connections with Research Findings**

The third key pain point we aim to address is the lack of soft-skill development. As one interview participant noted, “The cultivation of soft skills has been neglected in the alternative education pathway.”

- **The Recommendation**

Based on this finding, we propose Recommendation 3 – a Soft-Skill Development Tool. This tool functions as an AI chatbot that acts like an interviewer and asks users about their past experiences. This interaction provides an effective way to evaluate users’ soft skills. It also

encourages users to reflect on their learning experiences and think about how to articulate them in interviews. Users can use the tool to regularly review, assess, and practice their soft skills, ultimately strengthening their employability. This tool will be embedded as a plugin within the modular platform (Recommendation 1).

<Visual Mockup 3.  Soft Skill Development Tool Demo.mov >

- **Why It Matters**

Cousins, personal communication, November 13, 2024

Soft skills are critical competencies that employers evaluate during the hiring process (HR Professional 2, interview). Even after learners use the personalized modular learning platform and skill-tree roadmap to systematically acquire technical knowledge, a gap remains between learning and employment—specifically, the ability to demonstrate soft skills in interviews, such as problem-solving, communication, and teamwork. Recommendation 3 aims to help learners bridge this gap by reflecting on, assessing, and practicing how to present their previous learning experiences. This supports their ability to demonstrate soft skills effectively during interviews and increases their employability.

Employers commonly assess soft skills through case studies and scenario-based questions (HR Professional 2, interview). Answering these questions requires learners to recall and make sense of their technical knowledge and past experiences. However, most bootcamps or online courses do not provide structured opportunities for learners to reflect on, organize, or express what they have learned (Alternative-Pathway Learner 2, interview). As a result, alternative-pathway learners need a platform that supports reflection, evaluation, and practice. Simulating interview Q&A with an AI chatbot is an effective way to help users review and articulate their experiences.

- **Implementation and Constraints**

The AI chatbot's interview questions will be sourced from two areas: 1. web-scraped high-frequency interview questions across industries; 2. course-related questions, including knowledge-based, reflective, and scenario-based prompts.

Constraints include several stakeholder considerations: 1. The cost of calling AI model APIs is relatively high and may affect Instructure's profitability. 2. While using AI to simulate interviewers is effective for reviewing and evaluating users' soft skills, it has limited ability to directly improve those skills due to the absence of real-world practice environments. Soft skills cannot be developed solely through theoretical knowledge; they require applying concepts in practice, reflecting on the process, identifying issues, making adjustments, and continuously iterating. AI can only provide a simulated scenario, and it is difficult for it to fully replicate real interpersonal dynamics—such as human communication, negotiation, and emotional nuance—where current AI capabilities remain limited.

Conclusion Ping Yin **Completed** ▾

Main points summary

This project set out to understand how Instructure might better support alternative-pathway learners, and these individuals often face fragmented learning experiences, limited guidance, and a persistent employability gap. Through six interviews across alternative-pathway learners, HR professionals, and an Instructure employee, along with extensive background research and affinity wall analysis, our team identified three interrelated challenges: the need for structured and flexible learning access, the need for clear progress visibility, and the need for opportunities to practice and articulate soft skills. In response, we proposed a system of three integrated recommendations: a personalized modular learning platform, a skill-tree roadmap, and a soft-skill development tool.

Together, these recommendations operate as a coherent learning navigation ecosystem that supports users from knowledge acquisition to employability. The modular learning platform and skill-tree roadmap together form the structural backbone: enabling learners to acquire knowledge in flexible formats while understanding where they stand, what prerequisites matter, and what steps come next. soft-skill development tool, as a plugin layered on the whole system, shown as milestones on the roadmap. This tool is designed to help learners reflect on their experiences, practice articulating their competencies, and ultimately build confidence for real hiring situations. This end-to-end structure not only responds to learner needs uncovered in our research but also positions Instructure with a differentiated value proposition in the competitive EdTech landscape: supporting not just learning but also employability.

Limitations

Despite the promise of this integrated ecosystem, our research and recommendations also reveal several limitations. The personalized modular learning platform has significant potential impact and return, but its development requires substantial time and financial resources. It represents a “high-cost, high-risk, unclear ROI” long-term investment. The Soft-Skill Development Tool, although most aligned with Instructure’s strategy, involves higher AI-related costs, and AI-based training remains meaningfully different from real-world scenarios. Therefore, it can only serve as a supplementary tool for developing and practicing soft skills.

Absent stakeholder perspectives

Beyond these technical and strategic limitations, our research scope also reveals absent stakeholder perspectives.

We did not include Educational Policy Makers (government entities) in our interviews, even though they play a central role in shaping credentialing standards and the recognition of non-degree pathways. As a result, our project lacks policy-related insights that would help inform how credentials might be governed, implemented, or adopted in broader educational systems.

Further action

If additional resources were available, implementation should follow a phased, strategically aligned progression. Our team recommends beginning with the Soft-Skill Development Tool, as it best aligns with Instructure’s existing strategic priorities and requires comparatively lower development cost and time. As Melissa, Instructure’s Chief Academic Officer, noted, “Instructure is at the very beginning of developing technology to encourage soft skills.” Starting here would allow Instructure to pilot a career-readiness initiative, collect user feedback, and refine functionality before scaling into more ambitious offerings.

References

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Appendices:

Data Collection Sheet

- **Appendices:** Include supplementary materials, such as your anonymized data collection sheet. Do not share transcripts, interviewee names, or affinity wall contents with direct quotes. Some version of your data collection sheet (described above) should be submitted as an appendix.

AI Use Statement

Contribution & GenAI Use Statement

1) Contribution:

Each person should briefly describe the specific work each team member contributed to the project, such as research, analysis, writing, design, editing, coordination, or other responsibilities. The purpose is to clarify each group member's role, provide visibility into how you are applying course concepts, and ensure both individual and group efforts are recognized in the evaluation process.

Example: "I conducted two user interviews and summarized the findings for the group. I also edited the final draft for clarity and formatting."

2) GenAI Use:

Each person should explicitly describe their use of generative AI tools in this assignment. This should be the tool, the purpose, and the process.

Example: "I did not use any generative AI tools in this assignment" or "For this assignment, I employed Google Gemini to summarize three scholarly articles related to my research question. My goal was to more efficiently distill the main points from each article. I entered the abstract and asked Gemini to provide a concise summary. I compared these results to my own notes and used the summaries for reference, revising them with my own wording."

- **Wentao He** wentaohe@umich.edu: Designed the presentation's research process and research finding part, presented them, and wrote this part of the report.
- **Ping Yin** yinping@umich.edu: drew the user journey maps of each recommendation, presented the recommendation 3 and conclusion part in the final presentation, and authored the recommendation 3 and conclusion of the final report.
- **Xingyue Wang** wxingyue@umich.edu:
 1. **Contribution:** I contributed to all group meetings, I took the project manager role for two group assignments. I conducted two interviews as interviewer or notetaker. I participated in building affinity walls, brainstorming and presentations. For the final report and presentation, I designed and developed the whole structure, the user journey map, and the recommendation triangle of the

final presentation slide deck, presented the recommendations 1 and 2 in the final presentation, and authored the recommendation sections (recommendations 1 & 2) of the final report, support with the format and the citation of the final report.

2. **GenAI Use:** For the final report and presentation assignment, I used ChatGPT to proofread my section in order to be professional in the tone. I asked the ChatGPT to specifically point out the sentences it deems inappropriate and give suggestions without modifying the original sentence. After that, I compared the result to my own write-up and revised them with my own wording. Additionally, for the final presentation, I used Figma Make to generate the user journey map, I got the draft for the user journey map on google slides, my goal was to save time to build from scratches on Figma. After I got the generated map, Ping manually modified the contents referring to my draft. Figma make was used to complete the prototypes.
- **Layilah Nasser** layilah@umich.edu: Designed and developed key elements of the final presentation slide deck, including the layout HMW, Problem Statement, and built all the final prototypes, and authored the introduction and problem statement sections of the final report.

Presentation

PDF Link for the Presentation:  500-011-Team2-Presentation

Guidance on using Generative AI for this milestone

- You **MAY NOT** use generative AI tools to generate the insights/findings and critical analysis that form the bulk of the report.
- You **MAY** use generative AI tools to:
 - Aggregate and synthesize feedback for drafting the report or presentation
 - Draft outlines, bullet lists, or helping organize/structure your content
 - Edit for grammar and writing structure
 - Create inclusive images that help visualize presentation concepts or ideas

- Create charts, graphs, or other organization of data (ensure best practices in protecting identifying information when using AI for this purpose)
- Create the Executive Summary for your report, but it is your responsibility to ensure that the output adequately characterizes your report/project in sufficient detail.
- If you use generative AI, include an appendix with:
 - A brief description of what tools you used and how (~1 paragraph)
 - The prompt(s) you provided to the tool(s)
 - The output(s) generated
- We support responsible AI use that supports learning objectives. Failure to disclose use of AI may result in failing the assignment, and repeat offenses may be reported to UMSI academic integrity. **Caution:** Generative AI can create inaccurate or misleading content. You are responsible for everything you include from AI-generated material.

Submission Instructions

Contribution & GenAI Use Statement requirement

To support equitable collaboration and responsible tool use, each assignment must include a short statement at the end of the submission. This should cover both (1) your individual contributions to the project and (2) your individual and team use of generative AI tools.

- This is a **TEAM** assignment.
- There are multiple submission requirements for this assignment:
 - 1) Presentation: one person per team submit as a PDF:
Team#_M5_presentation.pdf
 - Note: you will also submit a link to your presentation prior to the last day of class per your instructor's instructions.
 - 2) Report: **You must send the report to your project partner, cc'ing your instructor.** This assignment will be considered complete following the delivery of the final report and all other supporting documentation being sent to the partner contact, with the instructor copied on the email.